

Technical Document #938: Database Systems - Comprehensive Overview

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Database replication copies data across multiple servers. It improves availability and provides fault tolerance.

Database connection pooling reuses database connections to reduce overhead. It improves performance for applications with many concurrent users.

Database monitoring tracks performance metrics and identifies bottlenecks. Query analysis and indexing strategies optimize database performance.

Relational databases organize data into tables with rows and columns. SQL is used to query and manipulate structured data.

Computer vision applications are becoming ubiquitous, from facial recognition to autonomous driving. Deep learning has enabled breakthroughs in visual understanding.

Voice assistants like Alexa and Siri use speech recognition and natural language understanding. They have become integral parts of smart home ecosystems.

Sustainability and environmental responsibility are becoming key considerations in technology design. Green computing aims to reduce the environmental impact of technology.

Data-driven decision making has become essential for modern businesses. Organizations that leverage data effectively gain a significant competitive advantage.

Key Takeaways:

- Sharding distributes data across multiple database servers.
- Database indexing improves query performance by creating data structures that allow faster data retrieval.
- Database caching stores frequently accessed data in memory.